Ordinary Differential Equations

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August 25, 2026

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1 Solutions of Differential Equations

Consider an nth-order differential equation of the form

$$y^{(n)} = f\left(t, y, y', \ldots, y^{(n-1)}\right),$$

where $y = y(t)$.

1.1 Definition of a Solution

A solution of this differential equation on an interval

$$\alpha < t < \beta$$

is a function

$$y = \phi(t)$$

such that

$$\phi', \phi'', \ldots, \phi^{(n)}$$

exist and

$$\phi^{(n)}(t) = f\left(t, \phi(t), \phi'(t), \ldots, \phi^{(n-1)}(t)\right)$$

for every t in the interval (α, β).

In other words, a solution is a function that satisfies the differential equation at every point in the interval under consideration.

1.2 Example

Consider the second-order differential equation

$$y'' + y = 0.$$

Equivalently,

$$y'' = -y.$$

Two solutions are

$$y_1(t) = \cos t$$

and

$$y_2(t) = \sin t.$$

For y_1,

$$y_1'(t) = -\sin t, \qquad y_1''(t) = -\cos t = -y_1(t).$$

For y_2,

$$y_2'(t) = \cos t, \qquad y_2''(t) = -\sin t = -y_2(t).$$

Thus both functions satisfy the differential equation for

$$-\infty < t < \infty.$$

Furthermore, because the differential equation is linear, any linear combination of these solutions is also a solution. Thus

$$y(t) = c_1 \cos t + c_2 \sin t$$

is a solution for any constants $c_1, c_2 \in \mathbb{R}$.

Therefore, this differential equation has infinitely many solutions.

The constants c_1 and c_2 allow us to select different solutions from this family.

2 First-Order Equations

A first-order differential equation has the general form

$$y' = f(t, y).$$

Two fundamental questions are:

1. Does a solution exist?
2. If a solution exists, is it unique?

These questions are central to the theory of differential equations.

There are several important types of first-order differential equations. In this course, we will consider:

1. linear equations,
2. separable equations,
3. exact equations.

3 First-Order Linear Equations

A first-order linear differential equation can be written in the form

$$y' + p(t)y = g(t),$$

where $p(t)$ and $g(t)$ are known functions.

Equivalently,

$$y' = -p(t)y + g(t).$$

When $p(t) = a$ and $g(t) = b$ are constants, we obtain

$$y' + ay = b.$$

This is called a first-order linear differential equation with constant coefficients.

For now, assume

$$a \neq 0.$$

3.1 The Integrating Factor

We want to solve

$$y' + ay = b.$$

The main idea is to multiply the entire equation by a carefully chosen function

$$\mu(t),$$

called an **integrating factor**.

Multiplying by $\mu(t)$ gives

$$\mu y' + a\mu y = \mu b.$$

We want the left-hand side to become the derivative of the product

$$\mu y.$$

By the product rule,

$$\frac{d}{dt}(\mu y) = \mu' y + \mu y'.$$

Therefore, we need

$$\mu' y + \mu y' = \mu y' + a\mu y.$$

Comparing the two expressions, we require

$$\mu' = a\mu.$$

Thus μ must satisfy the differential equation

$$\frac{d\mu}{dt} = a\mu.$$

Separating variables,

$$\frac{d\mu}{\mu} = a\,dt.$$

Integrating,

$$\int \frac{1}{\mu}\,d\mu = \int a\,dt,$$

so

$$\ln|\mu| = at + C.$$

Exponentiating,

$$|\mu| = Ce^{at}.$$

Since any nonzero constant multiple of an integrating factor is also an integrating factor, we may choose

$$\boxed{\mu(t) = e^{at}}.$$

Thus e^{at} is the integrating factor for

$$y' + ay = b.$$

3.2 Solving the Equation

Multiply

$$y' + ay = b$$

by the integrating factor

$$e^{at}.$$

We obtain

$$e^{at}y' + ae^{at}y = be^{at}.$$

By the product rule,

$$\frac{d}{dt}\left(e^{at}y\right) = be^{at}.$$

Integrating both sides,

$$\int \frac{d}{dt}\left(e^{at}y\right)\, dt = \int be^{at}\, dt.$$

Therefore,

$$e^{at}y = b\int e^{at}\, dt + C.$$

Since $a \neq 0$,

$$\int e^{at}\, dt = \frac{1}{a}e^{at}.$$

Hence

$$e^{at}y = \frac{b}{a}e^{at} + C.$$

Dividing by e^{at},

$$y = \frac{b}{a} + Ce^{-at}.$$

Therefore, the general solution is

$$\boxed{y(t) = \frac{b}{a} + Ce^{-at}}.$$

3.3 Initial Conditions

Suppose we are given the initial condition

$$y(0) = y_0.$$

Substituting $t = 0$ into the general solution,

$$y_0 = \frac{b}{a} + Ce^0.$$

Thus

$$y_0 = \frac{b}{a} + C,$$

so

$$C = y_0 - \frac{b}{a}.$$

Therefore, the solution to the initial-value problem

$$y' + ay = b, \qquad y(0) = y_0$$

is

$$\boxed{y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a}\right) e^{-at}}.$$

Notice that

$$\frac{b}{a}$$

is the equilibrium solution. The term

$$\left(y_0 - \frac{b}{a}\right) e^{-at}$$

describes how the solution differs from equilibrium based on the initial condition.

4 Example

Consider the initial-value problem

$$\frac{dy}{dt} + 2y = 3, \qquad y(0) = y_0.$$

Here,

$$a = 2, \qquad b = 3.$$

The integrating factor is

$$\mu(t) = e^{2t}.$$

Multiplying the differential equation by e^{2t},

$$e^{2t}\frac{dy}{dt} + 2e^{2t}y = 3e^{2t}.$$

Using the product rule,

$$\frac{d}{dt}\left(e^{2t}y\right) = 3e^{2t}.$$

Integrating,

$$e^{2t}y = \int 3e^{2t}\,dt = \frac{3}{2}e^{2t} + C.$$

Therefore,

$$y = \frac{3}{2} + Ce^{-2t}.$$

Applying the initial condition $y(0) = y_0$,

$$y_0 = \frac{3}{2} + C.$$

Thus

$$C = y_0 - \frac{3}{2}.$$

Hence the solution is

$$\boxed{y(t) = \frac{3}{2} + \left(y_0 - \frac{3}{2}\right)e^{-2t}}.$$

5 General First-Order Linear Equation

The same method works for the more general first-order linear equation

$$y' + p(t)y = g(t).$$

We seek an integrating factor $\mu(t)$ such that

$$\mu y' + \mu p(t) y$$

is the derivative of μy.

Since

$$\frac{d}{dt}(\mu y) = \mu' y + \mu y',$$

we require

$$\mu' = p(t)\mu.$$

Separating variables,

$$\frac{d\mu}{\mu} = p(t)\,dt.$$

Integrating,

$$\ln|\mu| = \int p(t)\,dt.$$

Thus we may choose

$$\boxed{\mu(t) = e^{\int p(t)\,dt}}.$$

Multiplying the original equation by $\mu(t)$,

$$\mu y' + \mu p(t) y = \mu g(t).$$

The left-hand side becomes

$$\frac{d}{dt}\left(\mu y\right),$$

so

$$\frac{d}{dt}\left(\mu y\right) = \mu g(t).$$

Integrating,

$$\mu(t) y(t) = \int \mu(t) g(t)\,dt + C.$$

Therefore,

$$\boxed{y(t) = \frac{\displaystyle\int \mu(t) g(t)\,dt + C}{\mu(t)}}$$

where

$$\boxed{\mu(t) = e^{\int p(t)\,dt}}.$$

This is the general integrating-factor method for first-order linear differential equations.

6 The Integrating Factor Method

Consider the first-order linear differential equation

$$y' + p(t) y = g(t).$$

We seek a function

$$\mu = \mu(t)$$

called an **integrating factor** such that, after multiplying the differential equation by $\mu(t)$, the left-hand side becomes the derivative of the product $\mu(t)y(t)$.

Multiplying the equation by $\mu(t)$,

$$\mu y' + \mu p(t) y = \mu g(t).$$

By the product rule,

$$\frac{d}{dt}(\mu y) = \mu' y + \mu y'.$$

Therefore, we want

$$\mu' y + \mu y' = \mu y' + \mu p(t) y.$$

This will occur provided that

$$\mu' = p(t)\mu.$$

Thus, the integrating factor must satisfy

$$\frac{d\mu}{dt} = p(t)\mu(t).$$

Separating variables,

$$\frac{d\mu}{\mu} = p(t)\, dt.$$

Integrating,

$$\int \frac{d\mu}{\mu} = \int p(t)\, dt,$$

so

$$\ln|\mu| = \int p(t)\, dt + C.$$

Exponentiating,

$$|\mu| = e^C e^{\int p(t)\, dt}.$$

Since a nonzero constant multiple of an integrating factor is also an integrating factor, we may choose

$$\boxed{\mu(t) = e^{\int p(t)\, dt}}.$$

6.1 Solving the Differential Equation

Multiplying

$$y' + p(t)y = g(t)$$

by the integrating factor gives

$$\mu y' + \mu p(t) y = \mu g(t).$$

By construction,

$$\frac{d}{dt}(\mu y) = \mu g(t).$$

Integrating,

$$\mu(t)y(t) = \int \mu(t) g(t)\, dt + C.$$

Therefore,

$$\boxed{y(t) = \frac{\displaystyle\int \mu(t)g(t)\,dt + C}{\mu(t)}}.$$

This gives the general solution of the first-order linear differential equation.

6.2 Initial-Value Problems

Suppose we are given the initial condition

$$y(t_0) = y_0.$$

It is often convenient to write the solution using a definite integral. Choose the integrating factor so that

$$\mu(t) = e^{\int_{t_0}^{t} p(s)\,ds}.$$

Then

$$\mu(t_0) = 1.$$

Since

$$\frac{d}{dt}\big(\mu(t)y(t)\big) = \mu(t)g(t),$$

integrating from t_0 to t gives

$$\mu(t)y(t) - \mu(t_0)y(t_0) = \int_{t_0}^{t} \mu(s)g(s)\,ds.$$

Using

$$\mu(t_0) = 1 \qquad \text{and} \qquad y(t_0) = y_0,$$

we obtain

$$\mu(t)y(t) = y_0 + \int_{t_0}^{t} \mu(s)g(s)\,ds.$$

Therefore, the solution to the initial-value problem is

$$\boxed{y(t) = \frac{y_0 + \displaystyle\int_{t_0}^{t} \mu(s)g(s)\,ds}{\mu(t)}},$$

where

$$\boxed{\mu(t) = e^{\int_{t_0}^{t} p(s)\,ds}}.$$

6.3 Example 1: A Simple Integrating Factor

Consider the differential equation

$$t^2\frac{dy}{dt} + 2ty = 0.$$

For $t \neq 0$, divide by t^2:

$$\frac{dy}{dt} + \frac{2}{t}y = 0.$$

Here,

$$p(t) = \frac{2}{t}.$$

The integrating factor is

$$\mu(t) = e^{\int \frac{2}{t}\,dt} = e^{2\ln|t|} = t^2.$$

Thus,

$$t^2\frac{dy}{dt} + 2ty = 0$$

can be written as

$$\frac{d}{dt}\left(t^2 y\right) = 0.$$

Integrating,

$$t^2 y = C.$$

Therefore,

$$\boxed{y(t) = \frac{C}{t^2}}, \qquad t \neq 0.$$

The solution is defined separately on intervals that do not contain $t = 0$, such as

$$(-\infty, 0) \qquad \text{or} \qquad (0, \infty).$$

6.4 Example 2: Solving an Initial-Value Problem

Consider the initial-value problem

$$t\frac{dy}{dt} + 2y = 4t^2, \qquad y(1) = 2.$$

First divide by t, assuming $t \neq 0$:

$$\frac{dy}{dt} + \frac{2}{t}y = 4t.$$

Thus,

$$p(t) = \frac{2}{t}, \qquad g(t) = 4t.$$

The integrating factor is

$$\mu(t) = e^{\int \frac{2}{t}\,dt}.$$

Since the initial condition is given at $t = 1$, we are working on the interval $t > 0$. Therefore,

$$\mu(t) = e^{2\ln t} = t^2.$$

Multiplying the differential equation by t^2,

$$t^2 \frac{dy}{dt} + 2ty = 4t^3.$$

The left-hand side is the derivative of $t^2 y$:

$$\frac{d}{dt}\left(t^2 y\right) = 4t^3.$$

Integrating,

$$t^2 y = \int 4t^3 \, dt = t^4 + C.$$

Therefore,

$$y = t^2 + \frac{C}{t^2}.$$

Apply the initial condition

$$y(1) = 2.$$

Then

$$2 = 1 + \frac{C}{1^2},$$

so

$$C = 1.$$

Hence the unique solution to the initial-value problem is

$$\boxed{y(t) = t^2 + \frac{1}{t^2}}, \qquad t > 0.$$

6.5 Behavior of the Solution

The solution

$$y(t) = t^2 + \frac{1}{t^2}$$

has the following limiting behavior.

As $t \to \infty$,

$$\lim_{t\to\infty} y(t) = \infty,$$

and more specifically,

$$\lim_{t\to\infty} \frac{y(t)}{t^2} = 1.$$

Thus, for large positive t,

$$y(t) \sim t^2.$$

As $t \to 0^+$,

$$\lim_{t \to 0^+} y(t) = \infty,$$

because

$$\frac{1}{t^2} \to \infty.$$

For example, at

$$t = 0.0001,$$

we have

$$y(0.0001) = (0.0001)^2 + \frac{1}{(0.0001)^2} = 10^{-8} + 10^8 \approx 100000000.$$

Thus, the solution grows extremely rapidly as t approaches 0 from the right.

6.6 Integral Curves

The family of solutions

$$y(t) = t^2 + \frac{C}{t^2}$$

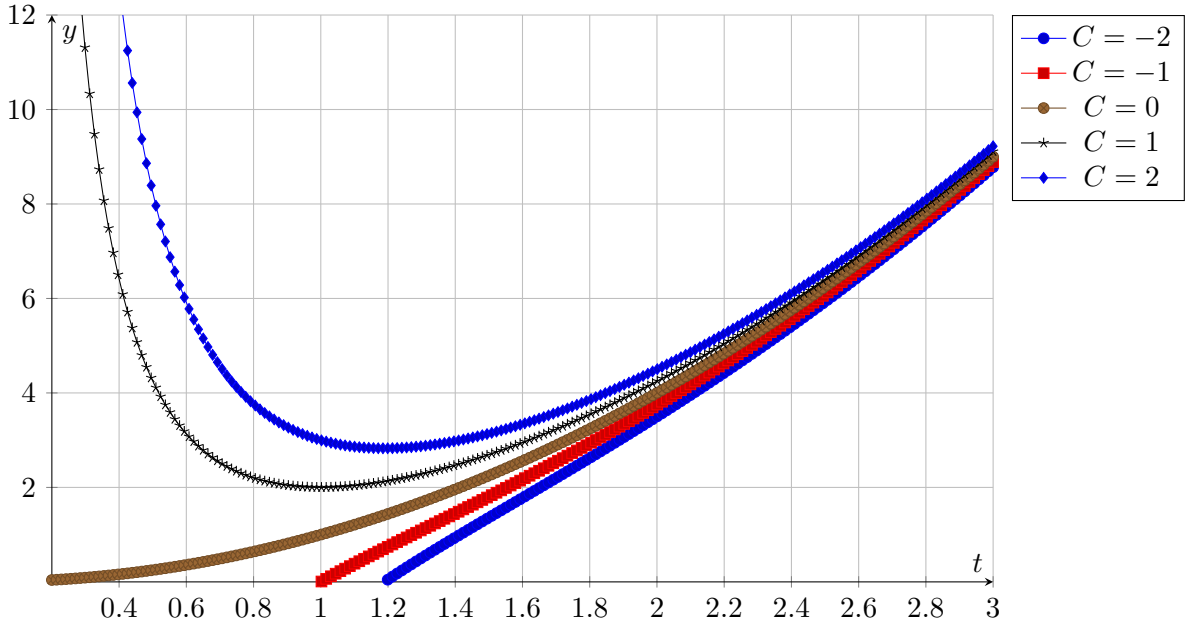
consists of integral curves of the differential equation

$$t\frac{dy}{dt} + 2y = 4t^2.$$

For the initial condition $y(1) = 2$, we obtain the particular solution

$$y(t) = t^2 + \frac{1}{t^2}.$$

The graph below shows several integral curves corresponding to different values of C.



The solution corresponding to the initial condition $y(1) = 2$ is the curve with

$$C = 1.$$